

Experiment -2

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Design a UI where users recall visual elements (e.g., icons or text chunks). Evaluate the effect of chunking on user memory.

FRAME 1:

INSTRUCTION PAGE:

Chunking Analysis of the Instruction Page Chunking is a cognitive strategy that breaks down information into smaller, manageable units, making it easier to process and retain. The Memory Recall Task instruction page effectively utilizes chunking in the following ways:

1. Clear and Sequential Numbering
 - The instructions are broken down into six steps, making it easier to follow
 - Each step presents one key action in a structured manner, reducing cognitive overload.
2. Logical Grouping of Information
 - Observation Phase (Steps 1-2) → Users learn about what they will see.
 - Memorization Strategy (Step 3) → Encourages users to focus on remembering items
 - Recall Phase (Steps 4-5) → Explains how users will recall information.
 - Key Performance Metric (Step 6) → Highlights that accuracy matters more than speed.
3. Visual Hierarchy and Design Elements
 - Bold, large title ("MEMORY RECALL TASK") → Grabs attention and clearly states the task.
 - Bullet points and spacing → Reduce clutter, enhancing readability.
 - Highlighted "START" button → Signals the next step, keeping navigation intuitive.

Time Constraint Reinforcement

 - Step 5 explicitly states, "You will have 5 seconds to view the items."
 - This reinforces expectations while subtly urging users to focus
4. Simplicity and Clarity
 - Instructions use short, direct sentences, avoiding unnecessary complexity.
 - The active voice makes it more engaging and action-oriented.

MEMORY RECALL TASK:

You will be shown several groups of icons or text.

After viewing, recall the items you remember.

You will have 5 seconds to view the items. Then, recall them in the next screen.



START

FRAME 2:

CHUNKING PHASE:

Analysis of the Memory Recall Task - Chunking Phase Screen This screen represents the Chunking Phase of a Memory Recall Task, where users observe and memorize different icons within a limited time. Below is a breakdown of its key components:

1. Purpose of the Screen

- This is the visual memory encoding phase, where users view and group items mentally before recalling them.
- The term "Chunking Phase" suggests that the game applies the chunking principle, which helps users remember items by categorizing them into meaningful groups.

2. Key Elements and UI Components

Header / Title – “Remember these items” (instruction text)

Main Container / Card – Holds all memory items

Item Rows (3 sections)

Fruits row

.Flowers row

Vehicles row

. **Icons / Emojis** – Visual memory items (uniform size & spacing)

Timer Display – “5 Seconds” countdown indicator

Background – Dark gradient for focus

Border / Frame – Decorative game-style frame

3 How the Chunking Phase Works

1. Users scan the grid and look for patterns or related items to create mental chunks.
2. The countdown timer limits observation time, forcing quick memory strategies.
3. Once time is up, users transition to the recall phase, where they must identify previously seen item

4. Cognitive and UX Benefits of Chunking

- Enhances short-term memory by allowing users to recall groups of information instead of individual elements.
- Reduces cognitive overload by helping users organize data efficiently.
- Improves pattern recognition, making recall easier and more accurate



RECALL PHASE:

Analysis of the Memory Recall Task - Selection Phase This screen represents the Selection Phase of a Memory Recall Task, where users recall and choose the items they remember from the previous Chunking Phase. Below is a breakdown of its components:

1. Purpose of the Screen

- This is the memory retrieval stage, where users select the items they remember seeing in the previous phase
- The goal is to test the effectiveness of chunking and short-term memory retention.

2.Key Elements & UI Components

- **Instruction Text** – “Enter the items you remember”
- **Input Fields (14)** – Text fields labeled Item 1 to Item 14
- **Form Layout** – Two-column structured list
- **Submit Button** – “SUBMIT” action button
- **Background** – Dark gradient for focus
- **Border / Frame** – Decorative page frame

3. How the Selection Phase Works

1. Users analyze the displayed icons and recall which they saw in the Chunking Phase.
2. They list down the remembered icons
3. Some icons are distractors, testing whether the user’s memory is accurate or if they mistakenly recall incorrect icons.
4. Clicking "SUBMIT" finalizes their choices, leading to a results or feedback screen

4. Cognitive and UX Benefits

- Tests memory accuracy by comparing user selections with previously displayed items.
- Incorporates distractors to evaluate how well users distinguish real vs. false memories.
- User-friendly interface with clear selection mechanics
- Gamified elements (bee character, bright colors) make the task engaging.

A screenshot of a memory recall task input form. The form has a dark background with a subtle, wavy, golden highlight effect. It is enclosed in a thick black border. At the top right, there is a white rectangular button with the word "SUBMIT" in a stylized, handwritten font. Below the button, the text "Enter the items you remember:" is displayed in a white, serif font. The form contains 14 input fields, arranged in two columns of seven. Each field is preceded by a label "Item 1:" through "Item 14:" in a white, bold, sans-serif font. The input fields themselves are thin white lines.

RESULT PAGE:

Analysis of the Memory Recall Task - Score & Feedback Screen

This screen represents the Score & Feedback Phase of the Memory Recall Task, where users receive their performance evaluation based on the selections made in the previous Recall Phase.

1. Purpose of the Screen

- Provides feedback on recall accuracy by showing the number of correct answers
- Allows users to decide their next action

2.Key Elements & UI Components

- **Result Text** – Score message (“10/14 items correctly”)
- **Feedback Message** – Positive confirmation
- **Primary Action Button** – “NEXT”
- **Background** – Dark + golden highlight effect
- **Layout Container** – Center-aligned content
- **Border / Frame** – Decorative outer frame

3. How This Phase Works

1. The game evaluates the user's selections from the Recall Phase
- . 2. It calculates the accuracy score (10/14 in this case) and displays it on a wooden board.
3. Users review their performance and choose their next action:

4. Cognitive & UX Benefits

- Instant feedback helps users track their memory performance.
- Multiple options (Continue, Restart, Exit) give users control over their learning experience
- . • Visual & gamified elements make the task engaging and less stressful



<https://www.figma.com/proto/EMycXmfgdQ1xsip1sWG73e/MEMORY-RECALL-TASK?node-id=6-6&t=8z8pxMdJTKp9sFCa-1>